L2b: Yield, Duration, Convexity, and the Yield Curve

CHEME 5660 Cornell University

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Risk. Trading involves risk and is generally not recommended for those with limited resources, experience, or a low risk tolerance. Past performance does not guarantee future success.

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Scenario

It is 6:30 AM at your trading desk. You are responsible for a $500 million portfolio of Treasury securities. The Federal Reserve meets today, and the rumor is a half-point move in the policy rate. The phone is already ringing.

By 2:00 PM you need an answer to one question: how much does this portfolio move?

That question has a conditional answer: it depends on the portfolio cash flows and on which yields move. Today we use the slope and curvature of the price-yield relationship to estimate the change.

Careful: a half-point move in the *policy rate* is not a half-point move in *your security*'s *yield*. We return to that.

Objectives for Today

Today we take the fixed-income price model from L2a and ask how its price responds to a change in the rate used to discount it.

Today's concepts

- Price versus yield: explain why the price of a fixed positive cash-flow stream decreases with yield and follows a convex curve.
- Duration and convexity: define both measures, estimate the local price effect of a yield change, and interpret their units.
- Yield-curve vocabulary: distinguish yield to maturity, par yield, spot rates, and the discount factors that price individual cash flows.

Companion notebook: L2b lecture notebook

Lectures and Examples

Lecture: CHEME-5660-L2b-Lecture-Yield-Duration-Convexity-Curve-Fall-2026.ipynb

The required example prices a coupon Treasury security and studies how its price responds to a change in yield and to a change in the coupon rate.

Example: CHEME-5660-L2b-Example-Treasury-Duration-Convexity-Fall-2026.ipynb

The longer theorem exercise and the term-structure notebooks stay available as optional extensions under advanced/.

Link: https://github.com/varnerlab/CHEME-5660-CourseRepository-Fall-2026.git

Quick review before we start: a safe payment can still have a risky price.

Concept Review: Are Treasury Securities Really Risk-Free?

No. Treasuries carry low default risk because they are obligations of the U.S. government, which is why the rate you declare is so often taken from them. That label is conditional. Two risks that remain matter today.

- Default risk: the issuer fails to pay. The United States has never repudiated its debt, though a 1979 operational failure did delay payment on some maturing bills. Other sovereigns have defaulted outright: Greece restructured its privately held debt in 2012, Russia defaulted in 1998.
- Interest-rate risk: if required yields change, the secondary-market value of a fixed promised cash flow changes. An institution that marks positions to market is affected even without a sale.

Today: interest-rate risk is the one we can compute. Its size depends on maturity, coupon, yield, and the shock.

Case Study: Interest Rate Risk and Silicon Valley Bank

SVB held a large portfolio of long-maturity fixed-rate government securities bought while rates were low. As the Federal Reserve raised rates through 2022 and 2023, the value of those securities fell.

- The losses were unrealized, so a hold-to-maturity story obscured them.
- In early March 2023 uninsured depositors withdrew funds. SVB announced a capital raise to cover realized bond losses, and the bank failed.
- The FDIC took control on March 10, 2023, with roughly $209 billion in assets: then the second largest failure of an insured bank on record, now the third.

Falling bond values and deposit withdrawals reinforced each other. Interest-rate risk was important, but it was not the only cause of the failure.

Overview: the collapse of Silicon Valley Bank, from PBS

Zero-Coupon Treasury Bill

Start with the simplest security: a bill that pays nothing until maturity and then returns its face value. This simple cash-flow pattern makes a bill a useful starting point.

Let $V_P > 0$ be the face value, $T > 0$ the remaining maturity in years, n the number of compounding intervals per year, and y the nominal annual yield used to discount the single promised payment. The per-period factor $1 + y/n$ is the growth factor over one interval, so the L2a discount factor is $\mathcal{D}_{T,0}^{-1}(y;n) = (1 + y/n)^{-nT}$. The price is given by:

$$V_B = \underbrace{(1 + y/n)^{-nT}}_{\text{discount}} V_P.$$

Standing assumption: $y > 0$ throughout this lecture, as it has been for the Treasury data we use. We will identify every result that depends on this assumption.

Deriving the Price Sensitivity

Three quantities could move the price: the yield y, the annual coupon rate c, and the maturity T. A bill's contract fixes $c = 0$, so $dc = 0$. Holding V_P and n fixed, only changes in y and T remain in the total differential:

$$dV_B = -\underbrace{V_P\left(1+y/n\right)^{-nT}\left(\frac{T}{1+y/n}\right)}_{V_{B,y}} dy \;-\; \underbrace{V_P\left(1+y/n\right)^{-nT}\left(n\log\left(1+y/n\right)\right)}_{V_{B,T}} dT.$$

Why there is no dc term: the bill's coupon rate is fixed and cannot change. In the coupon-bearing family of the next section, $\partial V_B/\partial c = (V_P/n)\sum_j \mathcal{D}_{j,0}^{-1}(y;n) > 0$, which stays strictly positive at $c = 0$. The derivative is not zero; the contract simply requires $dc = 0$.

What Does a Change in Maturity Mean?

A bill promises one payment, so its price depends on T only through the exponent nT. No intermediate payment can be added or removed, so $\partial V_B/\partial T$ is well defined here. But dT can describe two different comparisons:

- Changing the contractual maturity: comparing a 6-month bill with a 12-month bill compares two securities issued on different terms. Smooth for a bill, because only the exponent changes. Not smooth for a coupon security, where it adds or removes scheduled payments.
- The passage of time: holding one bill, calendar time t advances while remaining maturity falls, so $dT = -dt$ along the position. Smooth for a bill over the whole holding period. For a coupon security, smooth only between coupon dates.

Later: both distinctions return when we price coupon-bearing notes and bonds.

Yield and Maturity Sensitivities

Read $V_{B,y}$ and $V_{B,T}$ as magnitudes, not the derivatives themselves: the derivatives are their negatives, which is why the minus signs appear above. Under the standing assumption $y > 0$, both are positive:

$$V_{B,y} = \underbrace{V_P(1+y/n)^{-nT}}_{=V_B} \underbrace{\left(\frac{T}{1+y/n}\right)}_{>0}, \qquad V_{B,T} = \underbrace{V_P(1+y/n)^{-nT}}_{=V_B} \underbrace{(n\log(1+y/n))}_{=g_y},$$

where $g_y := n\log(1+y/n)$ is the continuously compounded equivalent of y from L2a. Given a well-posed price, $1 + y/n > 0$, the first magnitude is positive for any $T > 0$. The second needs $y > 0$: g_y carries the sign of y, so at $y = 0$ maturity stops mattering, and for $-n < y < 0$ a longer bill would be worth *more* and trade above par.

Careful: negative nominal yields have occurred in other sovereign markets. State the condition.

How Yield and Maturity Change a Bill's Price

Decreasing in the yield

Holding maturity fixed, a higher yield discounts the face value more heavily, so $V_B \downarrow$ as $y \uparrow$. Buy a 52-week bill at auction; if the yield moves from 2% to 3%, the secondary-market price falls below what you paid.

Decreasing in the time to maturity

Holding the yield fixed at a positive value, $V_B \downarrow$ as $T \uparrow$. But while you *hold* the bill, calendar time advances and the remaining maturity falls, so $dT < 0$: six months in, with the yield unchanged, the price is *higher* than you paid, because fewer periods remain to discount. At $y = 0$ exactly, every discount factor is one and the price does not move at all.

Optional: Bill Maturity Sensitivity

The required lecture now moves to coupon-bearing notes and bonds. The advanced bill notebook stays here in the sequence because it extends the maturity discussion we just completed.

Question: How does a zero-coupon bill's price sensitivity change as its remaining maturity changes?

Notebook: advanced/malkiel/, the Treasury bill maturity example.

Coupon-Bearing Treasury Notes and Bonds

Now let the security pay coupons. Let c be the annual coupon rate, so each payment is $C := (c/n)V_P$, and let $N := nT$ count the remaining payment dates, indexed by $j = 1, \ldots, N$. From L2a:

$$V_B = \mathcal{D}^{-1}_{N,0}(y;n)\, V_P + C \sum_{j=1}^{N} \mathcal{D}^{-1}_{j,0}(y;n), \qquad \mathcal{D}^{-1}_{j,0}(y;n) = (1 + y/n)^{-j}\,.$$

Holding that schedule fixed, the yield and coupon sensitivities are:

$$\frac{\partial V_B}{\partial y} = -\frac{1}{n(1+y/n)} \left[N V_P\, \mathcal{D}^{-1}_{N,0}(y;n) + C \sum_{j=1}^{N} j\, \mathcal{D}^{-1}_{j,0}(y;n) \right] < 0,$$

$$\frac{\partial V_B}{\partial c} = \frac{V_P}{n} \sum_{j=1}^{N} \mathcal{D}^{-1}_{j,0}(y;n) > 0.$$

Careful: N is a discrete schedule index. Differentiating a finite sum whose upper limit is nT with respect to a smooth T is not a valid total differential.

How Bond Features Change Its Price

- Decreasing in the yield: now *both* the coupons and the final principal are discounted more heavily, so $V_B \downarrow$ as $y \uparrow$.
- Increasing in the coupon rate, but not proportional to it: $\partial V_B / \partial c$ does not depend on c, and the discounted par payment survives even at $c = 0$. Doubling the coupon doubles the coupon stream, not the price.
- Maturity does not set the price direction by itself: a discount bond $(c < y)$ drifts up toward par as maturity shortens, a premium bond $(c > y)$ drifts down toward par, and a par bond $(c = y)$ sits at par on a coupon date.

Careful: Malkiel's maturity result is about *sensitivity to a yield change*, not about the price level.

Duration and Convexity

Derivatives tell us the direction of a price change. Risk management also needs its size in units that can be compared across positions.

We assume an option-free security: its promised cash flows do not change when yields change. Conventional noncallable Treasuries qualify; callable, inflation-indexed, and mortgage-backed securities do not.

Let N be the number of remaining periods and $X_j \geq 0$ the dollar cash flow at period j, with at least one $X_j > 0$. The same price formula covers both bills and coupon-bearing securities:

$$V_B(y) = \sum_{j=1}^{N} X_j \, \mathcal{D}_{j,0}^{-1}(y; n).$$

A note pays $X_j = C$ for $j < N$ and $X_N = C + V_P$; a bill pays $X_j = 0$ for $j < N$ and $X_N = V_P$.

Two Derivatives, One Shape

Differentiate the price twice with respect to the yield, holding the cash flows X_j and the schedule length N fixed:

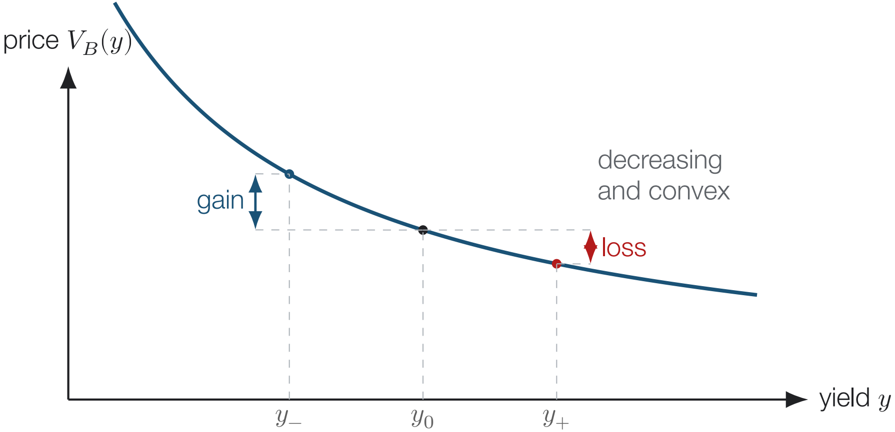
$$\frac{dV_B}{dy} = -\frac{1}{n}\sum_{j=1}^{N} j\,X_j\,(1+y/n)^{-j-1} < 0,$$

$$\frac{d^2V_B}{dy^2} = \frac{1}{n^2}\sum_{j=1}^{N} j(j+1)X_j\,(1+y/n)^{-j-2} > 0.$$

The strict signs follow from $1+y/n > 0$ and from at least one X_j being strictly positive: no term of either sum is negative, and at least one term of each is positive.

So the price-yield curve is decreasing and strictly convex. That second fact is called positive convexity, and it is what makes the price response asymmetric.

The Price-Yield Curve



Equal yield moves $y_\pm := y_0 \pm \Delta y$, for $\Delta y > 0$, from the same starting yield y_0. Convexity makes the gain exceed the loss.

The Two Risk Measures

Dividing each derivative by the price removes the size of the position and leaves two local risk measures. With $V_B(y) > 0$ the price at yield y, the modified duration and the convexity are given by:

$$D_{\mathrm{mod}}(y) := -\frac{1}{V_B(y)} \frac{dV_B}{dy}, \qquad K(y) := \frac{1}{V_B(y)} \frac{d^2V_B}{dy^2}.$$

Both are positive by the signs just established. With y a decimal per year, D_{mod} has units of years and K units of years squared.

Modified is not Macaulay. The Macaulay duration D_{mac} is the present-value-weighted average time to payment, and $D_{\mathrm{mod}} = D_{\mathrm{mac}}/(1 + y/n)$. For a bill, every dollar arrives at maturity, so $D_{\mathrm{mac}} = T$ and $D_{\mathrm{mod}} = T/(1 + y/n)$, which is exactly the ratio $V_{B,y}/V_B$ we derived earlier.

Duration Gives the Slope; Convexity Adds Curvature

A second-order Taylor expansion of $V_B(y)$ about the current yield, divided by the price, turns the two measures into a repricing estimate. For a yield change Δy, with $\Delta V_B := V_B(y + \Delta y) - V_B(y)$, the approximate fractional price change is given by:

$$\frac{\Delta V_B}{V_B} \approx \underbrace{-D_{\mathrm{mod}}\,\Delta y}_{\text{slope}} + \underbrace{\tfrac{1}{2} K (\Delta y)^2}_{\text{curvature}}.$$

Because $K > 0$, the curvature term is positive for a shock of *either* sign: it adds to the gain and subtracts from the loss.

The convention. Δy is a decimal change in the annual yield. One basis point is 10^{-4}, so a 50-basis-point shock is $\Delta y = 0.0050$.

Careful: a 50-basis-point move in the *policy rate* does not imply $\Delta y = 0.0050$ for a given security.

Estimate a Bond's Price Change

Take a note with $D_{\text{mod}} = 7.2$ years and $K = 65$ years^2, and shock its own yield by $+50$ basis points, so $\Delta y = 0.0050$:

- slope term: $-7.2 \times 0.0050 = -0.0360$
- curvature term: $\frac{1}{2} \times 65 \times (0.0050)^2 = +0.0008$
- estimate: a -3.52% price change, where the straight line alone would have said -3.60%

Both measures are local: they are derivatives at the current yield, so the estimate degrades as Δy grows. For the multiple-percentage-point moves of 2022 and 2023 that hurt SVB, the honest calculation is to reprice the whole schedule at the new yield, not to extrapolate from duration.

Connection to Malkiel's Bond Pricing Theorems

Two ground rules. Every theorem assumes an option-free security with fixed positive cash flows and holds the other contract terms fixed. And the maturity theorems compare bonds selling at par, $c = y$, measuring changes from that starting point. That restriction is not a technicality.

Substituting $C = (c/n)V_P$ and summing the geometric series puts the coupon bond in closed form. Differentiating that at fixed c, then setting $c = y$, gives the modified duration of a par bond:

$$\frac{V_B}{V_P} = \frac{c}{y} + \left(1 - \frac{c}{y}\right)(1 + y/n)^{-N} \qquad \Longrightarrow \qquad D_{\text{mod}}(N) = \frac{1 - (1 + y/n)^{-N}}{y}.$$

Order matters. Differentiate at fixed c, and only *then* set $c = y$. Setting $c = y$ first differentiates along the par locus, where the price is pinned at V_P, and returns zero. Assume $y > 0$ throughout.

Malkiel's Theorems 1, 2, and 3

- Theorem 1, prices move inversely to yields: this is the sign $dV_B/dy < 0$, or equivalently $D_{\text{mod}} > 0$. Every promised payment is positive, so raising y discounts each more heavily.
- Theorem 2, longer maturity increases yield sensitivity: $D_{\text{mod}}(N) = (1 - (1 + y/n)^{-N})/y$ is strictly increasing in N. A par bond starts at $V_B = V_P$ at every maturity, so the dollar sensitivity $-dV_B/dy = V_P D_{\text{mod}}(N)$ rises too: starting at par aligns the percentage and dollar rankings.
- Theorem 3, the gain diminishes: one extra period raises modified duration by $D_{\text{mod}}(N+1) - D_{\text{mod}}(N) = (1 + y/n)^{-N-1}/n$, and successive increments shrink by the constant factor $(1 + y/n)^{-1} < 1$, so D_{mod} climbs toward the ceiling $1/y$.

Note: $1/y$ is the modified duration of a *perpetuity*, a security that pays its coupon forever and never repays principal.

Malkiel's Theorems 4 and 5

- Theorem 4, price movements are asymmetric: a yield decrease raises the price by more than an equal increase lowers it, from the same starting yield. The reason is positive convexity. In the second-order estimate the slope term flips sign with the shock while the curvature term $\frac{1}{2}K(\Delta y)^2$ stays positive. Nonlinearity alone would not do it: a concave curve gives the opposite asymmetry.
- Theorem 5, higher-coupon bonds are less sensitive: this is a statement about *percentage* sensitivity, and it needs $N \geq 2$. A larger coupon moves present value to earlier dates, lowering the weighted average payment time and so lowering D_{mod}. In *dollars* it runs the other way: more cash flow means a larger $-dV_B/dy$.

Excluded edge: at $N = 1$ everything arrives on one date and $D_{\mathrm{mod}} = 1/\left[n(1 + y/n)\right]$, however large the coupon.

Why Malkiel Compares Bonds at Par

Outside the par comparison, percentage and dollar sensitivities can rank securities differently. For a zero-coupon bill, $D_{\text{mod}} = T/(1 + y/n)$ grows without bound, but the dollar sensitivity is a product of a growing and a shrinking factor:

$$-\frac{dV_B}{dy} = V_P \, \underbrace{\frac{T}{1 + y/n}}_{\text{growing}} \, \underbrace{(1 + y/n)^{-nT}}_{\text{shrinking}}.$$

So it rises, peaks, and falls. With $V_P = 100$, $n = 1$, $y = 5\%$ it peaks near $T \approx 20.5$ years at about 718 USD per unit of yield, then falls to about 541 by $T = 40$.

A 40-year bill is far more sensitive than a 20-year bill in percentage terms and *less* sensitive in dollars. Only the par comparison guarantees the two rankings agree.

Example: CHEME-5660-L2b-Example-Treasury-Duration-Convexity-Fall-2026.ipynb, which tests Theorems 4 and 5.

From One Yield to the Yield Curve

Yield to maturity is the single internal rate that reproduces one security's price from its promised cash flows. It is useful for quoting and comparison, but it is not the full term structure.

A spot rate applies to one payment date, so a spot curve supplies one discount factor per date. A par yield is the coupon rate that makes a hypothetical security trade at par under that curve. Writing $D(0, t_j)$ for the time-0 discount factor for maturity t_j, the curve-based price is:

$$V_B = \sum_{j=1}^{N} X_j \, D(0, t_j).$$

Curve level, slope, and curvature describe how discounting varies by maturity. They do not, by themselves, identify a unique forecast of future short rates.

Optional Advanced Material

Advanced 1: advanced/malkiel/
CHEME-5660-L2b-Advanced-Malkiel-TBill-Example-Fall-2026.ipynb

Advanced 2: advanced/yield_curve/
CHEME-5660-L3a-Lecture-YieldCurve-Data-Fall-2026.ipynb

Advanced 3: advanced/strips/
CHEME-5660-L3b-Lecture-STRIPS-Bonds-Fall-2026.ipynb

Optional; these preserve the deeper fixed-income material and are not prerequisites for L3a. The first compares Treasury bill yield sensitivity across maturities; the second (a lecture with three examples) covers yield to maturity, par and spot curves, and short-rate models; the third (a lecture and an issuer example) builds and prices Treasury STRIPS.

Notebooks: lectures/week-2/L2b/advanced, also linked from the lecture notebook.

Summary

This lecture developed the core tools used to describe and manage the interest-rate risk of fixed-income securities.

Key takeaways

- **Bond prices decrease as yields increase.** For a fixed stream of positive cash flows, the price-yield relationship is decreasing and convex.
- **Duration and convexity measure local price sensitivity.** Duration gives the first-order response to a yield change, and convexity corrects that straight line for curvature.
- **One yield is not a yield curve.** Yield to maturity summarizes one security with a single rate, while spot rates and discount factors value cash flows at specific maturities and par yields are implied by that term structure.

Together these form a practical fixed-income risk toolkit.

That’s a wrap. What are some of the interesting things we discussed today?